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Article

Projective Trace Residues in a Palatini Trace Sector: A Local Conditional Classification

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Abstract

Projective freedom is a standard feature of the trace sector in Palatini and metric-affine gravity. This note studies a narrower question: after a quotient-adapted trace reduction, does a projectively invariant trace residue remain inside a selected local operator class? In four dimensions the adopted law is the last-index projective transformation $\delta\Gamma^{\lambda}{}_{\mu\nu} = -\delta^{\lambda}{}_{\nu}\xi_{\mu}$, with compensator shift $\delta A_{\mu} = +3\xi_{\mu}$. The induced raw trace shifts are $\delta(T_{\mu}, Q_{\mu}, \widetilde{Q}_{\mu}, A_{\mu}) = (3, -8, -2, 3)\xi_{\mu}$. The three horizontal one-forms $q_{\mu} = Q_{\mu} + \frac{8}{3}A_{\mu}$, $u_{\mu} = \widetilde{Q}_{\mu} + \frac{2}{3}A_{\mu}$, and $t_{\mu} = T_{\mu} - A_{\mu}$ are coordinates on the quotient of the extended trace-plus-compensator space $(T, Q, \widetilde{Q}, A)$, together with the vertical coordinate $V_{\mu} = A_{\mu}$. They are not coordinates on the intrinsic quotient of the uncompensated triple $(T, Q, \widetilde{Q})$ alone. The one-form A_{μ} is a nondynamical spurionic reference along the projective fibre. The analysis uses only the compensated residue t_{μ}, in the restricted pure-trace density $f_1(\phi)R(\Gamma) + f_2(\phi)\, t_{\mu}t^{\mu} + f_3(\phi)\, \bar{\nabla}_{\mu}t^{\mu} + U(\phi)$. On a connected local region where $f_1 \neq 0$, elimination of q_{μ} and u_{μ} from the Palatini block is a Schur decoupling: $S_t = 2f_2$, $J_{t,\mathrm{eff}} = -\partial f_3$, and therefore $\Delta_2 = \Delta_3 = 0$, $\widehat{f_2} = f_2$, and $\widehat{f_3} = f_3$. No f_1-dependent t^2 term or $(\partial f_1)\cdot t$ source is induced. Strict removal means that the resulting residue expression vanishes identically, without solving for t_{μ} and without imposing an additional scalar constraint. On the relevant connected local branch this holds if and only if $f_2(\phi) \equiv 0$ and $f_3'(\phi) \equiv 0$. The statement is local, conditional, and confined to this compensated operator class.

Keywords: Palatini gravity; projective transformations; metric-affine geometry; trace residue; local classification

MSC: 53B05; 53C05; 83C05

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1. Introduction and Scope

Metric-affine and Palatini formulations treat the metric $g_{\mu\nu}$ and an independent affine connection $\Gamma^{\lambda}{}_{\mu\nu}$ as separate variables [2,4,7,8].

Projective freedom is a standard feature of this geometry: a shift of the connection along a pure trace direction changes representative data in the torsion and non-metricity traces while leaving a large part of the curvature scalar inert [4–6,8]. The question studied here is narrower than the general structure of metric-affine gravity. It asks whether, inside one selected local residue-operator class, a projectively invariant trace residue still enters the bulk variational equation after the trace sector has been written in quotient-adapted coordinates.

We work with the last-index projective law stated in Sec. 2. In this convention the trace-sector reduction is organized by an extended trace-plus-compensator quotient. The three horizontal coordinates used below are coordinates on the quotient of the extended trace-plus-compensator space $(T, Q, \widetilde{Q}, A)$ by the adopted projective direction. They should not be read as the intrinsic quotient of the uncompensated raw trace triple $(T, Q, \widetilde{Q})$ alone. The one-form A_{μ} is a nondynamical spurionic reference along the projective fibre; it permits the compensated horizontal coordinate

$$t_{\mu} = T_{\mu} - A_{\mu} \tag{1}$$

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but is not varied in this note, and it carries no stationarity equation. The variables q_μ and u_μ are invariant on the same extended quotient. They are retained only long enough to be eliminated from the curvature block.

The bulk functional is a restricted trace-residue action, written in Sec. 3. It is a chosen representative class inside the pure-trace distortion ansatz, selected so that the removal question for the compensated residue t_μ can be isolated. It is not presented as a complete two-derivative metric-affine basis, and the argument is not an unrestricted metric-affine theorem. The nontrivial payload is the Palatini Schur decoupling of Sec. 4.5: on a connected local region where $f_1 \neq 0$, eliminating q_μ and u_μ induces neither an f_1-dependent t^2 term nor a $(\partial f_1)\cdot t$ source, so $\Delta_2[f_1] = \Delta_3[f_1] = 0$. The strict-removal condition $f_2 \equiv 0$, with f_3 constant on that local branch, is the classification corollary of the resulting residue density. A separate problem, in which the spurionic reference is promoted to a varied auxiliary field, lies outside this note.

The result is local and conditional. It concerns one compensated residue chart, not global gluing, matter sources, or the stability of a gravitational background.

2. Projective Convention and the Trace Quotient

2.1. Adopted Projective Law

Spacetime dimension is $D = 4$, and the metric signature is $(-,+,+,+)$. Curvature, torsion, and non-metricity are defined in Appendix A. The projective transformation used throughout this paper is

$$\delta\Gamma^\lambda{}_{\mu\nu} = -\delta^\lambda{}_\nu\,\xi_\mu, \qquad \delta A_\mu = +3\xi_\mu, \qquad \delta g_{\mu\nu} = 0, \tag{2}$$

where ξ_μ is an arbitrary one-form. Index placement is part of the law. A different common convention shifts the first index, $\delta\Gamma^\lambda{}_{\mu\nu} = +\delta^\lambda{}_\mu\xi_\nu$. That first-index law is not used in this note. Appendix A records the short selection, inside a two-parameter trace family, that isolates (2).

Lemma 1 (Raw trace shifts)**.** *Under* (2) *in* $D = 4$,

$$\delta(T_\mu, Q_\mu, \widetilde{Q}_\mu, A_\mu) = (3, -8, -2, 3)\,\xi_\mu. \tag{3}$$

The Ricci tensor and the curvature scalar shift by

$$\delta R_{\mu\nu} = -2\partial_{[\mu}\xi_{\nu]}, \qquad \delta R = 0. \tag{4}$$

The proof is the direct contraction given in Appendix A. In particular, $\delta(T_\mu - A_\mu) = 0$.

2.2. Invariant Trace Coordinates

The raw shifts (3) determine three horizontal invariants and one vertical coordinate,

$$\begin{aligned} t_\mu &= T_\mu - A_\mu, & q_\mu &= Q_\mu + \frac{8}{3}A_\mu, \\ u_\mu &= \widetilde{Q}_\mu + \frac{2}{3}A_\mu, & V_\mu &= A_\mu. \end{aligned} \tag{5}$$

The inverse map is

$$\begin{aligned} T_\mu &= t_\mu + V_\mu, & Q_\mu &= q_\mu - \frac{8}{3}V_\mu, \\ \widetilde{Q}_\mu &= u_\mu - \frac{2}{3}V_\mu, & A_\mu &= V_\mu. \end{aligned} \tag{6}$$

Lemma 2 (Horizontal invariance)**.** *Under* (3), $\delta t_\mu = \delta q_\mu = \delta u_\mu = 0$ *and* $\delta V_\mu = 3\xi_\mu$. *Thus* t_μ, q_μ, *and* u_μ *are invariant coordinates on the quotient of the extended space* $(T, Q, \widetilde{Q}, A)$, *while* $V_\mu = A_\mu$ *runs along the projective fibre.*

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Proof. Substitute (3) into (5). Each horizontal combination is built so that the coefficient of ξ_μ cancels, and $\delta V_\mu = \delta A_\mu = 3\xi_\mu$. □

The three horizontal coordinates (q, u, t) are coordinates on the quotient of the extended trace-plus-compensator space $(T, Q, \widetilde{Q}, A)$ by the adopted projective direction. They should not be read as the intrinsic quotient of the uncompensated raw trace triple $(T, Q, \widetilde{Q})$ alone. In particular, $t_\mu = T_\mu - A_\mu$ is a compensated horizontal residue variable, not an additional intrinsic invariant of the uncompensated Palatini trace space. The coordinate t is the residue retained in the restricted action of Sec. 3. The variables q and u are invariant on the same extended quotient. Operators such as q^2, u^2, $q \cdot u$, $q \cdot t$, and $u \cdot t$ are invariant horizontal operators in that wider quotient. They lie outside the restricted residue class considered here. Projective symmetry does not remove them; the present action simply does not include them.

With A_μ held fixed as a spurionic reference, the map between $(T, Q, \widetilde{Q})$ and (q, u, t) at fixed V is linear and invertible on the horizontal trace coordinates. The residue variation used below is therefore a variation inside the selected horizontal chart of the extended quotient, after q and u have been eliminated. It is not supplemented by a stationarity equation for A_μ, and it is not a variation of an independent physical field in unrestricted metric-affine gravity. No equation $\delta S/\delta A_\mu = 0$ is imposed. After the change of coordinates (6), invariance of the restricted density along the adopted projective direction appears as independence of V_μ. That independence is used only to pass to the horizontal chart. Projective invariance below is understood spurionically under the simultaneous transformation of the affine connection and the nondynamical reference A_μ. It is not asserted to be a gauge redundancy acting on the dynamical connection while a fixed background A_μ is held unchanged.

3. Restricted Trace-Residue Action

Let ϕ be a scalar argument of the coefficient functions, and write $X \cdot Y = g^{\mu\nu} X_\mu Y_\nu$ and $X^2 = X \cdot X$. The bulk action studied in this paper is

$$S_{\rm res} = \int d^4x \, \sqrt{-g} \left[f_1(\phi)\, R(\Gamma) + f_2(\phi)\, t_\mu t^\mu + f_3(\phi)\, \bar{\nabla}_\mu t^\mu + U(\phi) \right]. \tag{7}$$

Here $R(\Gamma)$ is the curvature scalar of the independent connection, $\bar{\nabla}$ is the Levi-Civita derivative of $g_{\mu\nu}$, and $U(\phi)$ does not depend on the connection. The one-form t_μ is (1). The coefficient functions f_1, f_2, and f_3 are scalar functions of ϕ.

Equation (7) is a restricted trace-residue operator class: a selected local residue action, chosen to isolate the removal question for t_μ. The term $f_1 R(\Gamma)$ supplies the Palatini curvature block. The terms $f_2\, t^2$ and $f_3\, \bar{\nabla} \cdot t$ are the residue structures whose strict removability is classified below. No claim is made that (7) exhausts the projectively invariant scalar densities built from (q, u, t), nor that it is a complete two-derivative metric-affine basis. In particular, the invariant horizontal operators q^2, u^2, $q \cdot u$, $q \cdot t$, and $u \cdot t$ are absent from (7) by restriction of the class, not by a symmetry argument.

The quotient and residue-removal argument depends only on the stated projective trace-residue action.[1]

The metric equation and the scalar equation for ϕ are not part of the classification. Matter is not given an independent hypermomentum source in (7). The only variational expression used below is the bulk Euler–Lagrange expression for t_μ, after q_μ and u_μ have been eliminated on a connected local region where $f_1 \neq 0$. The reference A_μ is not varied.

[1] The PT-even scalar selection is retained only as provenance for how this restricted sector was isolated [1]. It is not a hypothesis of the Schur reduction or of the removal classification.

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4. Curvature Expansion and Palatini Elimination

4.1. Pure-Trace Curvature Identity

Equation (8) below is exact in the pure-trace distortion ansatz used here. It is not a complete decomposition of $R(\Gamma)$ for an unrestricted metric-affine connection. Axial torsion, tensor torsion, and traceless non-metricity are outside the coordinate domain of this note; the displayed expression is the trace block on which the restricted action (7) is defined. If a fully unrestricted connection is considered, (8) is that trace block after the non-trace irreducible pieces have been deleted from the domain, not a silent truncation of spectators that remain in the variational problem. Substituting the pure-trace distortion ansatz of Appendix A.2 into the connection-difference identity for $R(\Gamma)$ gives (8). Under the conventions of Appendix A, the pure-trace identity is

$$\begin{aligned} R(\Gamma) = R_{\mathrm{LC}} + \bar{\nabla}_\mu W^\mu - \frac{2}{3}T^2 - \frac{11}{72}Q^2 + \frac{1}{18}\widetilde{Q}^2 \\ - \frac{2}{3}T\cdot Q + \frac{2}{3}T\cdot\widetilde{Q} + \frac{2}{9}Q\cdot\widetilde{Q}, \end{aligned} \tag{8}$$

where the divergence current is the raw trace combination

$$W_\mu = -2T_\mu - Q_\mu + \widetilde{Q}_\mu. \tag{9}$$

There is no further quadratic trace structure in this identity. Its horizontal reduction is proved in Appendix B.

Proposition 1 (Vertical cancellation)**.** *After the substitution* (6)*, every term in* (8) *that contains V_μ cancels, and*

$$W_\mu = -2t_\mu - q_\mu + u_\mu. \tag{10}$$

The quadratic polynomial in (8) *reduces to the same polynomial evaluated on (t, q, u).*

Thus a pure vertical shift, $V_\mu \mapsto V_\mu + 3\xi_\mu$ with (q, u, t) held fixed, does not change $R(\Gamma)$. This is the curvature form of $\delta R = 0$ inside the trace sector. The restricted density (7) depends on the connection traces only through $R(\Gamma)$ and through t_μ, so the bulk integrand is independent of the spurionic coordinate V_μ.

4.2. Integration by Parts

Multiply (8) by f_1 and use the Levi-Civita identities

$$\sqrt{-g}\, f_1\, \bar{\nabla}_\mu W^\mu = -\sqrt{-g}\,(\partial_\mu f_1)\, W^\mu + \partial_\mu\big(\sqrt{-g}\, f_1 W^\mu\big), \tag{11}$$

$$\sqrt{-g}\, f_3\, \bar{\nabla}_\mu t^\mu = -\sqrt{-g}\,(\partial_\mu f_3)\, t^\mu + \partial_\mu\big(\sqrt{-g}\, f_3 t^\mu\big). \tag{12}$$

The currents generated by these rearrangements are

$$J_1^\mu = f_1 W^\mu, \qquad J_3^\mu = f_3 t^\mu. \tag{13}$$

Bulk densities that differ by the divergences $\partial_\mu(\sqrt{-g}\, J_1^\mu)$ and $\partial_\mu(\sqrt{-g}\, J_3^\mu)$ are identified in the bulk variational problem. This identification covers only the divergences produced by (11) and (12). It is not a boundary-charge analysis. Variations are taken with compact support in the local bulk patch, or with boundary data fixed so that the displayed divergences do not contribute to the Euler–Lagrange expressions.

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4.3. Horizontal Quadratic Form

Write $x = (q, u, t)^{\mathsf{T}}$ for the species triple of horizontal one-forms, with the quadratic pairing given by the metric contraction. After (11), (12), and Proposition 1, the affine bulk density, omitting $f_1 R_{\mathrm{LC}} + U(\phi)$ and omitting the admitted divergences, is

$$\mathcal{L}_{\mathrm{aff}} = \frac{1}{2}\, x^{\mathsf{T}} H_{\perp} x + J^{\mathsf{T}} x. \tag{14}$$

The species coefficient matrix and the source one-form are

$$H_{\perp} = f_1 H_0 + \mathrm{diag}(0, 0, 2f_2), \tag{15}$$

$$H_0 = \begin{pmatrix} -11/36 & 2/9 & -2/3 \\ 2/9 & 1/9 & 2/3 \\ -2/3 & 2/3 & -4/3 \end{pmatrix}, \tag{16}$$

$$J = \left(\partial f_1,\, -\partial f_1,\, 2\partial f_1 - \partial f_3\right)^{\mathsf{T}}. \tag{17}$$

Here ∂f_i denotes the gradient one-form of the scalar $f_i(\phi)$, and $J^{\mathsf{T}} x$ contracts each slot with the metric. The summand $\mathrm{diag}(0, 0, 2f_2)$ is the Hessian contribution of $f_2\, t^2$ in the normalization of (14). The matrix H_0 is the Hessian of the quadratic polynomial in (8) after the horizontal substitution. Both statements are checked in Appendix C. There is no V dependence in (14).

4.4. Elimination of q and u

Isolate the upper-left 2×2 block of $f_1 H_0$,

$$P = f_1 \begin{pmatrix} -11/36 & 2/9 \\ 2/9 & 1/9 \end{pmatrix}. \tag{18}$$

A direct computation, recorded in Appendix C, gives

$$\det P = -\frac{f_1^2}{12}, \qquad P^{-1} = f_1^{-1} \begin{pmatrix} -4/3 & 8/3 \\ 8/3 & 11/3 \end{pmatrix}, \tag{19}$$

on the locus $f_1 \neq 0$. Thus P is invertible if and only if $f_1 \neq 0$. At fixed residue t_μ, the Euler–Lagrange equation for (q, u) is algebraic. Its solution is

$$q_\mu = 4\,\frac{\partial_\mu f_1}{f_1} - \frac{8}{3} t_\mu, \qquad u_\mu = \frac{\partial_\mu f_1}{f_1} - \frac{2}{3} t_\mu. \tag{20}$$

4.5. Schur Data and Effective Residue Coefficients

Substitute (20) back into (14). The Schur complement of P in $H_{\perp}$, computed in Appendix C, is the scalar coefficient

$$S_t = 2f_2, \tag{21}$$

and the effective source one-form in the residue slot is

$$J_{t,\mathrm{eff}} = -\partial f_3. \tag{22}$$

The t-dependent bulk density collapses to

$$\mathcal{L}_t = f_2\, t_\mu t^\mu - (\partial_\mu f_3)\, t^\mu. \tag{23}$$

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The same reduction produces a remainder

$$\mathcal{L}_{\text{const}} = \frac{3\,(\partial f_1)^2}{2f_1}, \tag{24}$$

which is independent of t_μ, q_μ, and u_μ. It does not enter the residue equation.

Define the would-be induced residue couplings by

$$\widehat{f_2} = f_2 + \Delta_2[f_1], \qquad \widehat{f_3} = f_3 + \Delta_3[f_1]. \tag{25}$$

Comparison of (23) with the original residue terms in (7) gives

$$\Delta_2 = 0, \qquad \Delta_3 = 0, \tag{26}$$

and therefore

$$\widehat{f_2} = f_2, \qquad \widehat{f_3} = f_3. \tag{27}$$

Proposition 2 (Palatini Schur decoupling). *Let U be a connected local region on which $f_1(\phi) \neq 0$. Within the restricted compensated trace-residue class (7), eliminating q_μ and u_μ from the Palatini curvature block on U yields*

$$S_t = 2f_2, \qquad J_{t,\text{eff}} = -\partial f_3, \tag{28}$$

and hence there is no induced residue coupling:

$$\Delta_2[f_1] = 0, \qquad \Delta_3[f_1] = 0, \qquad \widehat{f_2} = f_2, \qquad \widehat{f_3} = f_3. \tag{29}$$

No f_1-dependent t^2 term or $(\partial f_1)\cdot t$ source is induced.

Proof. The block P, the determinant and inverse (19), and the horizontal solve (20) are the Palatini step on U. Appendix C evaluates the Schur complement $S_t = C - B^{\mathsf{T}}P^{-1}B$ and the effective source $J_t - B^{\mathsf{T}}P^{-1}J_N$. The f_1 piece of the quadratic t-coefficient cancels against $B^{\mathsf{T}}P^{-1}B$, and the ∂f_1 piece of the residue source cancels against $B^{\mathsf{T}}P^{-1}J_N$. The identities (21)–(27) follow. The remainder (24) does not depend on t_μ. □

The final removal condition is not the computational core by itself. Once $\mathcal{L}_t = f_2\,t^2 - (\partial f_3)\cdot t$ is obtained, strict removal is immediate. The nontrivial step is the prior Schur cancellation: the $f_1 R(\Gamma)$ block contains q, u, t mixing and a ∂f_1 source, yet eliminating q and u induces neither a t^2 contribution nor a $(\partial f_1)\cdot t$ residue source. Algebraic eliminability of an independent connection is a separate, already standard fact; the cancellation just stated is the content proved here.

5. Strict Residue-Removal Classification

By Proposition 2, the residue problem on U reduces to stationarity of $\mathcal{L}_t$. That stationarity condition is the residue equation

$$2f_2\,t^\mu - \partial^\mu f_3 = 0. \tag{30}$$

The distinction between solving this equation and removing the residue is the classification corollary.

Definition 1 (Strict residue removal). *Strict removal of t_μ from the bulk variational equation means that the Euler–Lagrange expression $2f_2\,t^\mu - \partial^\mu f_3$ vanishes identically as a functional expression, without solving for t_μ and without imposing an additional scalar constraint.*

Corollary 1 (Strict residue-removal classification). *Assume the projective law (2), the extended-quotient coordinates (5), and the restricted action (7). Let U be a connected local region on which $f_1(\phi) \neq 0$, so that*

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Proposition 2 applies and q_μ, u_μ are eliminated by (20). Then the remaining residue Euler–Lagrange expression on U is

$$2f_2\, t^\mu - \partial^\mu f_3. \tag{31}$$

This expression vanishes identically on U, in the sense of Definition 1, if and only if

$$f_2(\phi) \equiv 0, \qquad \mathrm{d}f_3 = 0 \tag{32}$$

as coefficient-function identities on the relevant connected local branch. Since ϕ is the only argument of the coefficient functions, this is equivalent to

$$f_2(\phi) \equiv 0, \qquad f_3'(\phi) \equiv 0 \tag{33}$$

on the connected interval of scalar-field values sampled by that branch. The condition is stronger than $\partial_\mu f_3 = 0$ on a single constant-ϕ solution: the latter would impose a solution restriction, not strict removal in the sense of Definition 1. On any branch in U where $f_2(\phi)$ is not identically zero, or where $\mathrm{d}f_3$ is not identically zero, the residue remains present in the restricted bulk variational problem. When $f_2 \neq 0$, the residue may still be solved algebraically,

$$t^\mu = \frac{\partial^\mu f_3}{2f_2}, \tag{34}$$

but that on-shell value is not strict removal. Crossing $f_1 = 0$ is a breakdown of the Palatini q/u Schur chart used here and is not classified in this note; no physical singularity is inferred from that algebraic exclusion alone.

Proof. By Proposition 2, the t-dependent bulk density on U is $\mathcal{L}_t = f_2\, t_\mu t^\mu - (\partial_\mu f_3)\, t^\mu$, up to the t-independent remainder $\mathcal{L}_{\mathrm{const}}$ and the admitted divergences generated by J_1 and J_3. Under the support condition of Sec. 4.2, those divergences do not contribute. Varying $\mathcal{L}_t$ with respect to t_μ produces (31).

Suppose (31) vanishes for every admissible t_μ and every scalar configuration whose values lie in the connected interval sampled by the branch, without an extra restriction on ϕ. Because $\partial_\mu f_3 = f_3'(\phi)\, \partial_\mu \phi$, the coefficient of t^μ and the scalar gradient must each vanish as functions of ϕ: $f_2(\phi) \equiv 0$ and $f_3'(\phi) \equiv 0$ on that interval. Vanishing of $\partial_\mu f_3$ only because one solution has constant ϕ does not give this identity.

Conversely, if $f_2(\phi) \equiv 0$ and $f_3'(\phi) \equiv 0$ on the sampled interval, then $\partial f_3 = 0$ for every such configuration and (31) is the zero expression. Definition 1 holds.

If $f_2(\phi)$ is not identically zero on the branch, then (31) retains the homogeneous piece $2f_2\, t^\mu$. Wherever $f_2 \neq 0$, one may solve (30) by (34). That step uses the residue equation; it does not make the Euler–Lagrange expression the zero expression before t_μ is substituted. If instead $f_2(\phi) \equiv 0$ while $f_3'(\phi) \not\equiv 0$, the expression reduces to $-\partial^\mu f_3$, which does not vanish as a coefficient identity. In both cases the residue remains in the restricted bulk problem, and strict removal fails. □

Remark 1 (Three local regimes)**.** *On the connected branch under discussion, the residue equation has three local regimes. If $f_2(\phi) \equiv 0$ and $\mathrm{d}f_3 = 0$ as coefficient identities, the expression is the identity $0 = 0$: this is the strict-removal branch. If $f_2 \neq 0$, then $t_\mu = \partial_\mu f_3/(2f_2)$ is an algebraic auxiliary solution, and it is not strict removal. If $f_2(\phi) \equiv 0$ but $\mathrm{d}f_3 \not\equiv 0$, variation with respect to t_μ produces the scalar consistency condition $\mathrm{d}f_3 = 0$; this multiplier branch is not removal, and it lies outside the strict-removal branch. These are regimes of one residue equation in the restricted class.*

The algebraic solution (34) may be substituted back into (20). The resulting values of q_μ and u_μ are on-shell coordinates in the restricted class. They are not used here to reconstruct a connection, and they are not needed for the removal statement.

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6. Control Branch, Limitations, and Scope

6.1. Pure Palatini Control Branch

Set $f_2 = 0$ and $f_3 = 0$, so that (7) reduces to the pure curvature term $f_1 R(\Gamma)$ plus $U(\phi)$. Then $\mathcal{L}_t = 0$. After elimination of q_μ and u_μ, the curvature block induces no t^2 term and no $(\partial f_1) \cdot t$ term. The only derivative piece generated by f_1 at this order is the t-independent density (24). Proposition 2 therefore says that the Palatini block induces no residue coupling: $\Delta_2 = \Delta_3 = 0$, $\widehat{f_2} = f_2$, and $\widehat{f_3} = f_3$. The pure branch sits inside the strict-removal locus (32) because both residue couplings are absent, not because a shifted combination of f_3 and f_1 has been set constant. In the present pure-trace compensated chart, the Schur reduction leaves no curvature-induced residue source.

The same conclusion covers every constant f_3, not only $f_3 = 0$. A constant f_3 changes the action by the admitted divergence generated by $J_3^\mu = f_3 t^\mu$, so it does not alter the bulk residue expression.

6.2. What the Classification Does Not Use

The hypotheses of Proposition 2 and Corollary 1 are the adopted projective law, the extended-quotient coordinates (q, u, t, V), the restricted action (7), and a connected local region on which $f_1 \neq 0$. A matter-sector closure condition, a preferred frame for additional fields, and any global extension beyond the local bulk expression are not used. The provenance footnote in Sec. 3 is not a step in the proof.

Where $f_2 = 0$ and $\partial f_3 \neq 0$ on U, the residue expression is $-\partial^\mu f_3$ and cannot be solved for t_μ. That branch is already counted in Corollary 1 as a failure of strict removal. It is not given a separate geometric taxonomy here.

6.3. Relation to Connection-Elimination Literature

Algebraic elimination of an independent connection is standard in Palatini and scalar–metric-affine gravity [9–14]. Projective invariance belongs to that structure, and integrating out the connection is a known route to an equivalent metric theory. This paper does not rediscover that eliminability. The specific contribution is narrower: in a compensated trace-residue chart and in the restricted action (7), the $f_1 R(\Gamma)$ curvature block induces no residue coupling under the Schur elimination of q and u, and the strict-removal classification follows from that. The phrase is classical and algebraic: no quantum or renormalization-group statement is intended.

The same distinction applies to inflationary integrations of torsion and non-metricity. Andrei, Dioguardi, Iosifidis, Järv, Racioppi, and Saal couple a scalar to the non-Riemannian Ricci scalar and to divergences of the torsion and non-metricity vectors, solve the connection equations, and integrate those vectors out in an inflationary setting [15]. This note does not build an inflationary or Einstein-frame model. It focuses on the compensated residue $t = T - A$ and on strict functional removal inside a restricted local chart of the extended quotient. The overlap is the structural use of divergence couplings and of connection or trace elimination; the question and the scope are different.

6.4. Scope

The classification is local: coefficient identities are identities of the local functions f_i on the connected region U, and the elimination of q_μ and u_μ is an algebraic step of the Schur chart on that region. It is conditional on the restricted class (7) and on the pure-trace domain of Sec. 4.1. Adding any of the invariant operators q^2, u^2, $q \cdot u$, $q \cdot t$, or $u \cdot t$ changes the horizontal quadratic form and takes the model outside Proposition 2. The result is not an unrestricted metric-affine theorem, and it is not a statement about physical stability, phenomenology, or the propagation of tensor modes.

The compensator A_μ remains a nondynamical spurionic reference along the projective fibre. No equation $\delta S / \delta A_\mu = 0$ is imposed. Analyses in which A_μ is varied as an auxiliary field, and in which the vertical fibre is treated as part of an enlarged stationarity problem, are outside this note.

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7. Conclusions

Under the last-index law $\delta\Gamma^{\lambda}{}_{\mu\nu} = -\delta^{\lambda}{}_{\nu}\xi_{\mu}$, the four-dimensional trace shifts are $\delta(T, Q, \widetilde{Q}, A) = (3, -8, -2, 3)\xi$. The coordinates (q, u, t) belong to the quotient of the extended trace-plus-compensator space, with A a nondynamical spurionic reference and $t = T - A$ the compensated residue. Inside the restricted pure-trace action, Proposition 2 is the payload: on a connected local region where $f_1 \neq 0$, eliminating q and u yields $\Delta_2 = \Delta_3 = 0$, so $\widehat{f_2} = f_2$ and $\widehat{f_3} = f_3$. Corollary 1 then states that strict removal holds on that branch if and only if $f_2(\phi) \equiv 0$ and $f_3'(\phi) \equiv 0$ on the scalar interval it samples. On every other branch of this class the residue remains in the bulk variational problem, even where $f_2 \neq 0$ permits the algebraic solution $t = \partial f_3/(2f_2)$.

The statement stops at that local chart. It does not vary A_{μ}, and it does not extend the pure-trace block to an unrestricted connection.

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Appendix A. Conventions and Trace Shifts

This appendix fixes the tensor conventions and proves Lemma 1. Greek indices are spacetime indices. Symmetrization and antisymmetrization include the conventional factor $\frac{1}{2}$. The Levi-Civita derivative of $g_{\mu\nu}$ is written $\bar{\nabla}$. The independent connection defines its own derivative ∇, used only to introduce non-metricity.

Appendix A.1. Curvature, Torsion, and Non-Metricity

The curvature of Γ is

$$R^{\lambda}{}_{\rho\mu\nu}(\Gamma) = \partial_{\mu}\Gamma^{\lambda}{}_{\nu\rho} - \partial_{\nu}\Gamma^{\lambda}{}_{\mu\rho} + \Gamma^{\lambda}{}_{\mu\sigma}\Gamma^{\sigma}{}_{\nu\rho} - \Gamma^{\lambda}{}_{\nu\sigma}\Gamma^{\sigma}{}_{\mu\rho}, \tag{A1}$$

with Ricci contraction $R_{\mu\nu} = R^{\lambda}{}_{\mu\lambda\nu}$ and scalar $R = g^{\mu\nu}R_{\mu\nu}$. Torsion and its trace are

$$T^{\lambda}{}_{\mu\nu} = 2\Gamma^{\lambda}{}_{[\mu\nu]}, \qquad T_{\mu} = T^{\lambda}{}_{\lambda\mu}. \tag{A2}$$

Non-metricity and its two traces are

$$Q_{\lambda\mu\nu} = -\nabla_{\lambda}g_{\mu\nu}, \qquad Q_{\lambda} = g^{\mu\nu}Q_{\lambda\mu\nu}, \qquad \widetilde{Q}_{\nu} = g^{\mu\lambda}Q_{\lambda\mu\nu}, \tag{A3}$$

where $\nabla_{\lambda}g_{\mu\nu} = \partial_{\lambda}g_{\mu\nu} - \Gamma^{\rho}{}_{\lambda\mu}g_{\rho\nu} - \Gamma^{\rho}{}_{\lambda\nu}g_{\mu\rho}$. The present reduction is defined on the pure trace/vector domain. Axial torsion, tensor torsion, and traceless non-metricity are not varied spectator fields in the classification; they are outside the domain of the restricted action.

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Appendix A.2. Pure-Trace Distortion Ansatz

Write the independent connection as the Levi-Civita connection of $g_{\mu\nu}$ plus a distortion,

$$\Gamma^{\lambda}{}_{\mu\nu} = \left\{{}^{\lambda}{}_{\mu\nu}\right\} + N^{\lambda}{}_{\mu\nu}. \tag{A4}$$

In the pure trace/vector domain the distortion is

$$\begin{aligned} N^{\lambda}{}_{\mu\nu} = {} & \frac{1}{3}\left(\delta^{\lambda}{}_{\mu}T_{\nu} - g_{\mu\nu}T^{\lambda}\right) \\ & + \frac{1}{36}\left(5\delta^{\lambda}{}_{\mu}Q_{\nu} + 5\delta^{\lambda}{}_{\nu}Q_{\mu} - 7g_{\mu\nu}Q^{\lambda}\right) \\ & + \frac{1}{18}\left(-\delta^{\lambda}{}_{\mu}\widetilde{Q}_{\nu} - \delta^{\lambda}{}_{\nu}\widetilde{Q}_{\mu} + 5g_{\mu\nu}\widetilde{Q}^{\lambda}\right). \end{aligned} \tag{A5}$$

Contracting the torsion of N with (A2) returns the torsion trace that was used as input,

$$N^{\lambda}{}_{\lambda\mu} - N^{\lambda}{}_{\mu\lambda} = T_{\mu}. \tag{A6}$$

The same contractions of the non-metricity (A3), in $D = 4$, return Q_{μ} and $\widetilde{Q}_{\mu}$. No axial torsion, tensor torsion, or traceless non-metricity is included. This is the pure-trace/vector domain of this note: those irreducible pieces are outside the ansatz, not spectator fields carried along and left unvaried.

Appendix A.3. Shifts Under the Adopted Law

Proof of Lemma 1. Write $\delta\Gamma^{\lambda}{}_{\alpha\beta} = -\delta^{\lambda}{}_{\beta}\xi_{\alpha}$. Then

$$\delta T_{\mu} = \delta\Gamma^{\lambda}{}_{\lambda\mu} - \delta\Gamma^{\lambda}{}_{\mu\lambda} = -\xi_{\mu} - (-4\xi_{\mu}) = 3\xi_{\mu}. \tag{A7}$$

The same shift changes the metric derivative by $\delta(\nabla_{\lambda}g_{\mu\nu}) = 2\xi_{\lambda}g_{\mu\nu}$, so

$$\delta Q_{\lambda\mu\nu} = -2\xi_{\lambda}g_{\mu\nu}. \tag{A8}$$

Contraction gives $\delta Q_{\lambda} = -8\xi_{\lambda}$ and $\delta\widetilde{Q}_{\nu} = -2\xi_{\nu}$. Together with $\delta A_{\mu} = 3\xi_{\mu}$, this is (3).

For the curvature, the linear piece is

$$\delta_{\mathrm{lin}}R^{\lambda}{}_{\rho\mu\nu} = \partial_{\mu}(\delta\Gamma^{\lambda}{}_{\nu\rho}) - \partial_{\nu}(\delta\Gamma^{\lambda}{}_{\mu\rho}) = -\delta^{\lambda}{}_{\rho}\, 2\partial_{[\mu}\xi_{\nu]}. \tag{A9}$$

The quadratic remainder built from two insertions of $\delta\Gamma$ cancels:

$$\delta\Gamma^{\lambda}{}_{\mu\sigma}\,\delta\Gamma^{\sigma}{}_{\nu\rho} - \delta\Gamma^{\lambda}{}_{\nu\sigma}\,\delta\Gamma^{\sigma}{}_{\mu\rho} = \delta^{\lambda}{}_{\rho}\,\xi_{\mu}\xi_{\nu} - \delta^{\lambda}{}_{\rho}\,\xi_{\nu}\xi_{\mu} = 0. \tag{A10}$$

The Ricci contraction of (A9) is $\delta R_{\mu\nu} = -2\partial_{[\mu}\xi_{\nu]}$. The metric trace of that antisymmetric tensor vanishes, so $\delta R = 0$. □

Appendix A.4. First-Index Convention, Not Used Here

The two-parameter family containing the usual first-index and last-index writings is

$$\delta\Gamma^{\lambda}{}_{\mu\nu} = a\,\delta^{\lambda}{}_{\mu}\xi_{\nu} + b\,\delta^{\lambda}{}_{\nu}\xi_{\mu}, \qquad \delta A_{\mu} = +3\xi_{\mu}. \tag{A11}$$

The torsion-trace contraction of (A11) is $\delta\Gamma^{\lambda}{}_{\lambda\mu} = (4a + b)\xi_{\mu}$ and $\delta\Gamma^{\lambda}{}_{\mu\lambda} = (a + 4b)\xi_{\mu}$, so $\delta T_{\mu} = 3(a - b)\xi_{\mu}$. The non-metricity contraction is $\delta Q_{\lambda\mu\nu} = a(\xi_{\mu}g_{\lambda\nu} + \xi_{\nu}g_{\lambda\mu}) + 2b\,\xi_{\lambda}g_{\mu\nu}$, and its traces are $\delta Q_{\mu} = (2a + 8b)\xi_{\mu}$ and $\delta\widetilde{Q}_{\mu} = (5a + 2b)\xi_{\mu}$. The linear curvature contraction is

$$\delta_{\mathrm{lin}}R_{\alpha\beta} = b\,\partial_{\alpha}\xi_{\beta} + (-3a - b)\,\partial_{\beta}\xi_{\alpha}. \tag{A12}$$

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Imposing $\delta T_\mu = 3\xi_\mu$ and $\delta_{\rm lin} R_{\mu\nu} = -2\partial_{[\mu}\xi_{\nu]}$ forces $a - b = 1$, $b = -1$, and $-3a - b = 1$, hence $a = 0$ and $b = -1$. That is the adopted law (2), for which (A10) shows that the quadratic remainder vanishes as well. The first-index pair $(a, b) = (1, 0)$ produces $\delta(T, Q, \widetilde{Q}) = (3, 2, 5)\xi$ and does not reproduce (4). It is a different convention, recorded here only to mark the boundary of the present targets.

Appendix B. Trace Curvature Expansion

This appendix proves Proposition 1. For the curvature convention (A1), the connection-difference identity is

$$R(\Gamma) - R_{\rm LC} = \bar{\nabla}_\mu \big(N^\mu{}_\lambda{}^\lambda - N^\lambda{}_\lambda{}^\mu\big) + \mathcal{Q}(N), \tag{A13}$$

where $\mathcal{Q}(N)$ denotes the quadratic contractions of the distortion. Substitution of the pure-trace ansatz (A5) gives

$$W_\mu = N_\mu{}^\lambda{}_\lambda - N^\lambda{}_{\lambda\mu} = -2T_\mu - Q_\mu + \widetilde{Q}_\mu \tag{A14}$$

and the six quadratic coefficients displayed in (8). That substitution is the trace block of the restricted action. It is not a complete decomposition of $R(\Gamma)$ for an unrestricted metric-affine connection. The vertical cancellation below starts from this W_μ and from those six coefficients. Substitute $T = t + V$, $Q = q - \frac{8}{3}V$, and $\widetilde{Q} = u - \frac{2}{3}V$.

For the current,

$$\begin{aligned} W &= -2(t + V) - \Big(q - \frac{8}{3}V\Big) + \Big(u - \frac{2}{3}V\Big) \\ &= -2t - q + u + \Big(-2 + \frac{8}{3} - \frac{2}{3}\Big)V \\ &= -2t - q + u. \end{aligned} \tag{A15}$$

The coefficient of V vanishes, which is (10).

For the quadratic polynomial, expand each monomial and collect the coefficients of V^2, $t \cdot V$, $q \cdot V$, and $u \cdot V$. The six contributions are as follows. From $-\frac{2}{3}T^2$: $-\frac{2}{3}$, $-\frac{4}{3}$, 0, 0. From $-\frac{11}{72}Q^2$: $-\frac{88}{81}$, 0, $\frac{22}{27}$, 0. From $\frac{1}{18}\widetilde{Q}^2$: $\frac{2}{81}$, 0, 0, $-\frac{2}{27}$. From $-\frac{2}{3}T \cdot Q$: $\frac{16}{9}$, $\frac{16}{9}$, $-\frac{2}{3}$, 0. From $\frac{2}{3}T \cdot \widetilde{Q}$: $-\frac{4}{9}$, $-\frac{4}{9}$, 0, $\frac{2}{3}$. From $\frac{2}{9}Q \cdot \widetilde{Q}$: $\frac{32}{81}$, 0, $-\frac{4}{27}$, $-\frac{16}{27}$. Each column sums to zero:

$$V^2: \quad -\frac{2}{3} - \frac{88}{81} + \frac{2}{81} + \frac{16}{9} - \frac{4}{9} + \frac{32}{81} = 0, \tag{A16}$$

$$t \cdot V: \quad -\frac{4}{3} + \frac{16}{9} - \frac{4}{9} = 0, \tag{A17}$$

$$q \cdot V: \quad \frac{22}{27} - \frac{2}{3} - \frac{4}{27} = 0, \tag{A18}$$

$$u \cdot V: \quad -\frac{2}{27} + \frac{2}{3} - \frac{16}{27} = 0. \tag{A19}$$

The substituted polynomial is therefore identical to the original polynomial evaluated on (t, q, u),

$$\begin{aligned} \mathcal{Q}(t, q, u) = &-\frac{2}{3}t^2 - \frac{11}{72}q^2 + \frac{1}{18}u^2 \\ &- \frac{2}{3}t \cdot q + \frac{2}{3}t \cdot u + \frac{2}{9}q \cdot u. \end{aligned} \tag{A20}$$

Combined with (A15), this is Proposition 1. The identity $\delta R = 0$ for a pure vertical shift is the same cancellation read in the opposite direction: the shifts (3) move $(T, Q, \widetilde{Q})$ along V at fixed (q, u, t), and both W and $\mathcal{Q}$ are invariant.

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Appendix C. Palatini Block and Schur Calculation

Appendix C.1. Source and Hessian

After Proposition 1, the linear bulk terms produced by (11) and (12) are

$$-(\partial f_1)\cdot W-(\partial f_3)\cdot t=(\partial f_1)\cdot(2t+q-u)-(\partial f_3)\cdot t. \tag{A21}$$

Read as a row acting on $x=(q,u,t)^{\mathsf{T}}$, this is J in (17).

The Hessian of $\mathcal{Q}(q,u,t)$ in the species slots, with the normalization $\mathcal{Q}=\frac{1}{2}x^{\mathsf{T}}H_0x$, has entries

$$\begin{aligned}\partial_q\partial_q\mathcal{Q}&=-\frac{11}{36}, & \partial_u\partial_u\mathcal{Q}&=\frac{1}{9}, & \partial_t\partial_t\mathcal{Q}&=-\frac{4}{3},\\ \partial_q\partial_u\mathcal{Q}&=\frac{2}{9}, & \partial_q\partial_t\mathcal{Q}&=-\frac{2}{3}, & \partial_u\partial_t\mathcal{Q}&=\frac{2}{3}.\end{aligned} \tag{A22}$$

These are the entries of H_0 in (16). Adding the second derivative of $f_2\,t^2$, which contributes $2f_2$ in the t slot of $\frac{1}{2}x^{\mathsf{T}}H_\perp x$, gives (15).

Appendix C.2. Determinant and Inverse

Let

$$M=\begin{pmatrix}-11/36 & 2/9\\ 2/9 & 1/9\end{pmatrix},\qquad P=f_1M. \tag{A23}$$

Then

$$\det M=\left(-\frac{11}{36}\right)\left(\frac{1}{9}\right)-\left(\frac{2}{9}\right)^2=-\frac{11}{324}-\frac{4}{81}=-\frac{27}{324}=-\frac{1}{12}, \tag{A24}$$

so $\det P=-f_1^2/12$. The inverse of M is

$$M^{-1}=\begin{pmatrix}-4/3 & 8/3\\ 8/3 & 11/3\end{pmatrix}. \tag{A25}$$

Indeed,

$$\begin{aligned}\left(-\frac{11}{36}\right)\left(-\frac{4}{3}\right)+\left(\frac{2}{9}\right)\left(\frac{8}{3}\right)&=1,\\ \left(-\frac{11}{36}\right)\left(\frac{8}{3}\right)+\left(\frac{2}{9}\right)\left(\frac{11}{3}\right)&=0,\end{aligned} \tag{A26}$$

and the remaining two products follow by symmetry of M. Hence $P^{-1}=f_1^{-1}M^{-1}$ on $f_1\neq 0$, which is (19).

Appendix C.3. Solution for q and u

Partition $H_\perp$ as

$$H_\perp=\begin{pmatrix}P & B\\ B^{\mathsf{T}} & C\end{pmatrix},\qquad B=f_1\begin{pmatrix}-2/3\\ 2/3\end{pmatrix},\qquad C=-\frac{4}{3}f_1+2f_2, \tag{A27}$$

and write $y=(q,u)^{\mathsf{T}}$ together with $J_N=(\partial f_1,-\partial f_1)^{\mathsf{T}}$. The cross term $y^{\mathsf{T}}(Bt)$ in $\frac{1}{2}x^{\mathsf{T}}H_\perp x$ uses the off-diagonal block once. Stationarity in y at fixed t is

$$Py+Bt+J_N=0. \tag{A28}$$

Therefore $y=-P^{-1}(Bt+J_N)$. Set $r=\partial f_1/f_1$ and $n=(1,-1)^{\mathsf{T}}$, so $J_N=(\partial f_1)\,n$. Then

$$M^{-1}n=\begin{pmatrix}-4\\ -1\end{pmatrix},\qquad -P^{-1}J_N=\begin{pmatrix}4r\\ r\end{pmatrix}. \tag{A29}$$

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The mixing column gives

$$M^{-1}\begin{pmatrix}-2/3\\2/3\end{pmatrix}=\begin{pmatrix}8/3\\2/3\end{pmatrix},\qquad -P^{-1}(Bt)=-\begin{pmatrix}(8/3)\,t\\(2/3)\,t\end{pmatrix}. \tag{A30}$$

Adding (A29) and (A30) gives (20).

Appendix C.4. Schur Complement

The Schur complement in the t slot is $S_t = C - B^{\mathsf{T}}P^{-1}B$. Using the second product in (A30),

$$B^{\mathsf{T}}P^{-1}B = f_1\begin{pmatrix}-2/3 & 2/3\end{pmatrix}\begin{pmatrix}8/3\\2/3\end{pmatrix} = f_1\Big(-\frac{16}{9}+\frac{4}{9}\Big) = f_1\Big(-\frac{4}{3}\Big). \tag{A31}$$

Hence

$$S_t = \Big(-\frac{4}{3}f_1+2f_2\Big)-\Big(-\frac{4}{3}f_1\Big) = 2f_2. \tag{A32}$$

The effective source is likewise

$$J_{t,\mathrm{eff}} = J_t - B^{\mathsf{T}}P^{-1}J_N = (2\partial f_1-\partial f_3)-2\partial f_1 = -\partial f_3, \tag{A33}$$

where $J_t = 2\partial f_1 - \partial f_3$ is the last slot of (17). The source contraction is $B^{\mathsf{T}}P^{-1}J_N = (\partial f_1)\,(-2/3)(-4) + (\partial f_1)\,(2/3)(-1) = 2\partial f_1$, by (A29). The quadratic-plus-linear density in t is therefore

$$\frac{1}{2}S_t\,t^2 + J_{t,\mathrm{eff}}\cdot t = f_2\,t^2 - (\partial f_3)\cdot t, \tag{A34}$$

which is (23). The t-independent remainder is

$$\mathcal{L}_{\mathrm{const}} = -\frac{1}{2}J_N^{\mathsf{T}}P^{-1}J_N = \frac{3\,(\partial f_1)^2}{2f_1}, \tag{A35}$$

because $J_N^{\mathsf{T}}M^{-1}J_N = -3(\partial f_1)^2$ and one power of f_1^{-1} remains. No term linear in t survives from f_1, and the f_1 piece of C is cancelled by $B^{\mathsf{T}}P^{-1}B$. Consequently $\Delta_2[f_1]=0$ and $\Delta_3[f_1]=0$, so $\widehat{f_2}=f_2$ and $\widehat{f_3}=f_3$.

Varying (A34) with respect to t_μ multiplies the quadratic term by two and raises the index on ∂f_3, producing $2f_2\,t^\mu - \partial^\mu f_3$. This is the expression used in Corollary 1.

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